using short-wave length UHF radio waves from fixed and mobile devices, and building personal area networks (PANs).

Infrared Radio Link: IR wireless is the use of wireless technology in devices or systems that convey data through infrared (IR) radiation. IR wireless is used for short- and medium-range communications and control.

Satellite Link: Satellite is a wireless technology having significant importance across the globe. They have found wide spread use in specialized situations.The devices using satellite technology to communicate directly with the orbiting satellite through radio signals. This allows users to stay connected virtually from anywhere on the earth. Portable satellite phones and modems have powerful broadcast feature and reception hardware than the cellular devices due to the increased range.The satellite communication consists of a space segment and a ground segment. When the signal is sent to the satellite through a device, the satellite amplifies the signal and sent it back to the receiver antenna which is located on the earth's surface. The ground segment consists of a transmitter, receiver and the space segment, which is the satellite itself.

Wi-Fi: Wi-Fi is a low-cost wireless communication technology. A WiFi setup consists of a wireless router which serves a communication hub, linking portable device with an internet connection.This network facilitates connection of many devices depending on the router configuration. These networks are limited in range due to the low power transmission, allowing the user to connect only in the close proximity.



Fig.6.8. Wireless Technologies

Do You Know?

- E-mail was in existence even before the origin of the web.
- More than 80% of the emails sent daily are spams.
- CAPTCHA – Completely Automated Public Turning test to tell Computers and Humans Apart
- 40 – 55% of Wikipedia vandalism is caught by a single computer program with 90% accuracy.

6.11. Cybercrimes:

Any criminal activity that uses a computer either as an instrumentality target or a means to spread further crimes that comes within the scope of cyber crime. A generalized definition of cybercrime may be "unlawful acts wherein the computer is either a tool or target of both".

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